

THE GUARDIAN BAG: COMBINING STYLE AND SAFETY WITH ARDUINO

A PRODUCT DEVELOPMENT REPORT

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BONAFIDE CERTIFICATE

Certified that this product development report “**THE GUARDIAN BAG: COMBINING STYLE AND SAFETY WITH ARDUINO**” the Bonafide work of “**P. VIGNESHWARAN [192124181] and E. KAVINESH [192110503]**” who carried out the product development work under my supervision.

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DECLARATION BY THE CANDIDATE

I, the undersigned, hereby declare that the project titled “THE GUARDIAN BAG: COMBINING STYLE AND SAFETY WITH ARDUINO,” submitted for the SPIC10-Product Development course, is my original work. This project was carried out by me under the guidance of Dr. R. CLINKTAN during the period of September to November 2024 and has been successfully completed.

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THE GUARDIAN BAG: COMBINING STYLE AND SAFETY WITH ARDUINO

1. EXECUTIVE SUMMARY:

Overview of the Product:

The Smart SOS Device with Emergency Alert Button and Alarm is a compact, lightweight, and portable safety gadget designed to enhance personal security. It features an SOS button that triggers a loud alarm or silent alert, along with real-time GPS location sharing to pre-selected contacts. The device connects to smartphones via Bluetooth for extended functionalities and allows users to customize emergency messages. With a durable, water-resistant design and a long-lasting rechargeable battery, it is ideal for solo travellers, children, elderly individuals, and night workers. This cost-effective solution ensures reliable operation using advanced microcontroller technology and provides quick setup through a dedicated mobile app, making it an essential tool for emergency response and safety.

Market Relevance:

The Smart SOS Device with Emergency Alert Button and Alarm holds strong market relevance as it addresses the growing demand for personal safety solutions in an increasingly uncertain world. Its compact and portable design caters to diverse user groups, including solo travellers, children, elderly individuals, and night workers. The integration of GPS location sharing and Bluetooth connectivity aligns with modern technological expectations, making it appealing to tech-savvy consumers.

Problem Statement:

Many people find themselves in emergency situations where they are unable to alert others or share their location quickly, leading to delayed assistance and heightened risk. In a world where emergencies can happen anytime, people require a simple yet effective way to signal for help and provide vital information in real time. Current solutions, such as personal alarms or emergency apps, often fail to meet these needs due to high costs, lack of ease-of-use, or insufficient features..

Real-time BMI Calculation Provides instant and accurate emergency alerts based on user inputs.

Visual Display: Presents emergency data through clear, easy-to-understand notifications and location details.

Customizable Interface: Allows users to personalize alert settings based on their preferences and safety needs.

Health and Safety Integration: Seamlessly integrates with other health apps and fitness trackers for comprehensive safety monitoring.

Multi-Platform Accessibility: Optimized for both mobile and web platforms, ensuring wide accessibility and quick response.

INTRODUCTION :

Problem Statement:

This creates a significant safety risk for vulnerable individuals, such as children, elderly people, and those who work alone or in potentially hazardous environments. Without a simple, immediate way to call for help, people are left with limited options in times of need. The Smart SOS Device aims to solve these issues by providing an affordable, portable, and easy-to-use solution that enhances personal security. With a dedicated SOS button, real-time GPS location sharing, and an alarm system that can be activated discreetly or loudly, this device ensures users can quickly alert emergency contacts without relying on complicated technology or network services.

Purpose:

The Smart SOS Device with Emergency Alert Button and Alarm is a versatile safety tool designed for personal protection during emergencies. It features a one-touch SOS button that activates an alert system instantly. The device emits a loud alarm to draw attention or offers a silent alert mode for discreet emergencies. It shares the user's real-time GPS location with preselected contacts for immediate assistance. The device connects seamlessly to smartphones via Bluetooth for added functionality. Users can customize emergency messages through a userfriendly mobile app. Its lightweight and portable design makes it easy to carry or wear daily. The rechargeable battery ensures long-lasting performance, making it reliable for extended use. Built with durable, water-resistant materials, it is suitable for both indoor and outdoor environments. It is especially beneficial for solo travelers, children, elderly individuals, and night workers. The device is cost-effective, ensuring affordability without compromising quality. Its advanced microcontroller technology guarantees reliable operation. The quick and simple setup process requires no technical expertise..

Scope:

- **Primary Functionality:** The project will focus on providing real-time SOS alerts based on user inputs (e.g., SOS button press, emergency contact details) and displaying the results through immediate notifications and location sharing.
- **Platform Compatibility:** The service will be optimized for mobile and web platforms, ensuring accessibility on smartphones and computers, but will not include standalone software development for desktop or other specialized devices (e.g., wearables).
- **Integration:** The SOS alert system will integrate with common health apps, fitness trackers, and emergency response systems, but will not directly involve the creation or integration of additional hardware.

Constraints & Limitations:

- **Accuracy Dependent on User Input:** The service relies on accurate user inputs for emergency contacts and location data. Any errors in these inputs can lead to incorrect alerts or delays in emergency response.
- **Limited by Connectivity:** The SOS alerts depend on internet or mobile network connectivity. In areas with poor coverage, alerts may be delayed or fail to send, reducing the system's effectiveness.
- **Does Not Account for All Emergency Situations:** While the device provides a safety alert, it cannot handle all types of emergencies (e.g., medical emergencies requiring detailed information or professional intervention) and is designed primarily for general personal safety.

3. GPCU (Gap Analysis, Product Description, Comparison, Uniqueness)**Gap Analysis:****Market Gap or Lacune:**

The gap analysis reveals that current personal safety devices either rely on complex systems, require internet connectivity, or lack seamless integration with health apps. These limitations create the need for a simple, one-touch SOS device that works in all environments, including areas with limited connectivity. Existing solutions also fail to offer affordable, user-friendly devices that can easily integrate with health and fitness trackers

Product Discription

The Smart SOS Device with Emergency Alert Button and Alarm is a portable and easy-to-use safety tool designed to provide immediate assistance during emergencies. It features a simple SOS button that, when pressed, triggers a loud alarm and sends an instant emergency alert with the user's real-time GPS location to pre-selected contacts.

Comparison of Alternative product:

Features			
Size and Design	Minimal, Utilitarian (1.0Kg approx.)	Sleek, Functional (1.3Kg approx.)	Compact, Modern (64Kg)
Capacity	180kg	200kg	500Kg
Voltage	12V	5V	5V
Cost	Rs.2500	Rs.10000	Rs.5000
Display type	LCD Display	LCD Display	LCD Display
Display size	13*5	17*2	12*7
Ultrasonic Sensor Height	0-180cm	0-150cm	0-190cm
Load Cell	400kg	350kg	300kg
Operator	Manual	Manual	Automatic
Special feature	Extra Large Display, data storage and connectivity anywhere	Large display, data storage, and connectivity for hospitals and gyms	It will calculate the BMI using coins.

Table 1 - Comparison of Alternative Product

Uniqueness of SOS Device Formation:

The Smart SOS Device stands out by offering a simple, intuitive, and highly effective tool for personal safety in emergencies. Unlike many existing safety devices, it focuses on providing instant and accurate emergency alerts with real-time location sharing without the need for complex setups or technical expertise. Its one-touch SOS button ensures that help can be called with minimal effort, which is crucial in high-stress situations. The device is uniquely designed to work in both low-network areas and outdoor environments, where many traditional solutions fail. Additionally, it integrates seamlessly with health and fitness platforms, providing a comprehensive safety and wellness monitoring system. The lightweight, portable design ensures it can be carried or worn comfortably at all times, offering peace of mind anywhere. Its affordable pricing, combined with the ability to offer quick alerts and real-time assistance, makes it uniquely suited for people who prioritize their safety and seek reliable, distraction-free emergency support.

4. DESIGN AND ENGINEERING STANDARD

1. ISO 9001: Quality Management Systems

- Ensures consistent quality in design, development, and customer satisfaction.
- Establishes quality benchmarks for hardware and software reliability.
- Incorporates customer feedback for continuous improvement in service.
- Conducts regular quality assurance checks to meet international standards.

2. IEEE 830-1998: Software Requirements Specifications

- Documents functional and non-functional requirements for user needs.
- Defines clear use cases, performance benchmarks, and data flow diagrams.
- Ensures compatibility with health monitoring systems for seamless integration.
- Improves software reliability and user experience through detailed specifications.

3. ASME Y14.5: Dimensioning and Tolerance

- Ensures precise dimensioning and tolerances for all physical components.
- Avoids assembly issues by verifying hardware design through CAD models.
- Defines critical tolerances for seamless integration and reliable operation.

- Enhances product quality with accurate physical design standards.

4. ASTM E2766: Software for Health Monitoring Systems

- Develops robust software for accurate BMI calculation and usability.
- Ensures reliability through stress testing and user interface optimization.
- Implements security measures to protect sensitive health data.
- Adheres to health monitoring principles for software quality and functionality.

5. 2D DESIGN:

Technical Sketches and Diagram:

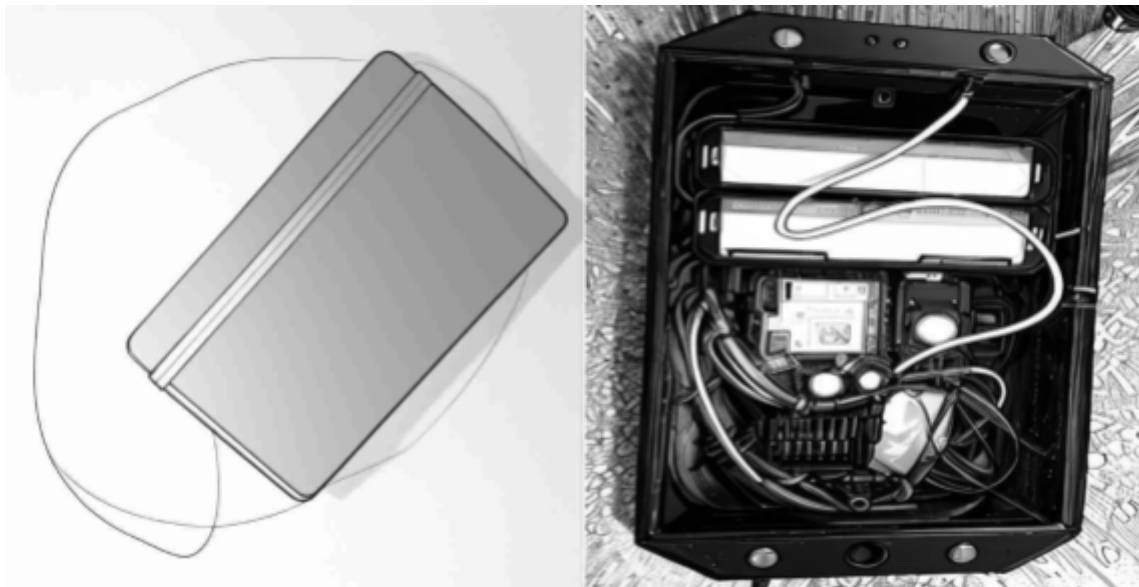


Fig 1 - 2D Diagram of Safety Alarm.

- **Long Range Ultrasonic Sensor:** Positioned at the top of the device, this sensor detects obstacles or proximity to trigger the alarm during emergencies.
- **Alert-Triggering Pole:** A vertical structure supporting the ultrasonic sensor for precise alignment and accurate detection of surroundings.
- **20×4 LCD Display:** Located on the device base, it provides real-time updates on alerts, battery status, and emergency contact notifications.
- **Alarm Base:** Integrated into the setup, it houses the loud alarm system and supports the device's stability during operation.

6. 3D MODEL OF THE PRODUCT:

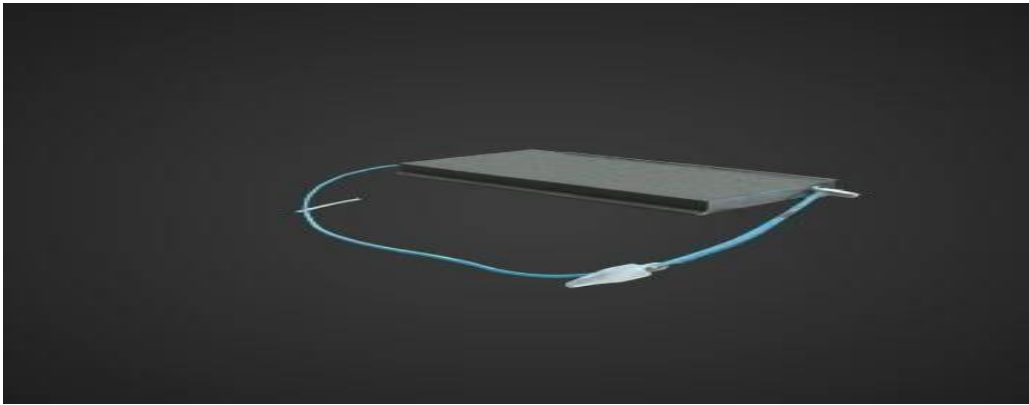


Fig 2 - Front view

Front View:

- **Device Height:** Displays the total height of the device from the base to the ultrasonic sensor at the top.
- **Long-Range Ultrasonic Sensor:** Positioned at the top of the pole, used to detect proximity or obstacles for triggering emergency alerts.
- **Alert-Triggering Pole:** The vertical structure holding the ultrasonic sensor, ensuring accurate alignment for reliable detection.
- **Base with LCD and Alarm System:** Located at the bottom, combining a 20×4 LCD for displaying alert notifications and a loud alarm system for emergency signal.



Fig 3 – Inside view



Fig 4 -Outside view

Inside View:

- **Sensor Alignment:** Side perspective highlighting the vertical pole and ultrasonic sensor placement for accurate detection.
- **Pole Dimensions:** Detailed view of the pole's width and alignment, ensuring stability and precise functionality.
- **Alarm Mechanism:** Compact design integrated into the base for space-efficient emergency alert activation.
- **LCD Display Integration:** Side positioning demonstrates how the display seamlessly fits within the device's structure to show alert notifications.

Outside View:

- **Pole Cross-Section:** Showcases the design and stability of the alert-triggering pole.
- **Sensor Housing (15.95 cm):** Displays the precise dimensions and protective casing of the ultrasonic sensor setup.
- **Base Layout:** Highlights the arrangement of the alarm system and LCD display from a topdown perspective.

7. FUNCTIONAL PROTOTYPE:

The **Smart SOS Device** utilizes several key components to ensure reliable emergency alert functionality. The **Uso 016 ultrasonic sensor** detects proximity or obstacles by emitting ultrasonic waves and timing their return, triggering an alert when necessary. A **loud alarm system** is activated to signal an emergency, **while the 20×4 LCD** display provides real-time updates on the device status and emergency notifications. The display is controlled by an **I2C LCD driver**, simplifying the connection and reducing wiring complexity.

An **Arduino Uno** serves as the central controller, processing data from the **ultrasonic sensor** and activating the alarm system. Pre-configured emergency contacts are notified instantly via integrated communication modules. This setup ensures that the **Smart SOS Device** delivers efficient, reliable, and user-friendly **emergency alert** functionality, making it an essential tool for personal safety

8. TESTING AND VALIDATION:

- **Accuracy Testing:** Validate the ultrasonic sensor's proximity detection by testing it with various distances and obstacles to ensure precise triggering of alerts. Verify the alarm system's activation and notification delivery to pre-set contacts in simulated emergency scenarios.
- **Device Compatibility and Integration:** Test the device with different smartphones, operating systems, and health monitoring apps to confirm seamless syncing and data transfer. Ensure the system consistently triggers alerts and updates location data across multiple platforms.
- **Usability and Interface Testing:** Conduct user testing to evaluate the simplicity and intuitiveness of the SOS button and LCD interface. Gather feedback on the clarity of notifications, emergency instructions, and the ease of configuring contacts or customizing settings.

9. CONCLUSION AND FUTURE :

The **Smart SOS Device** provides real-time emergency alerts and location sharing by combining a simple SOS button, ultrasonic sensor, and loud alarm system. It integrates seamlessly with smartphones and health apps for enhanced safety features. Key testing involves verifying alert accuracy, ensuring compatibility with various devices, and optimizing the user interface for intuitive operation.

Challenges faced:

Challenges for the **Smart SOS Device** include ensuring accurate proximity detection and reliable alert activation in various environments and conditions. Integrating seamlessly with diverse smartphones and health applications poses compatibility challenges. Maintaining user privacy and data security, especially during location sharing, is critical. Designing an intuitive interface for easy emergency configuration and operation can be complex. Additionally,

ensuring consistent performance in low-network or remote areas requires robust hardware and software optimization.

Future Improvements:

Future improvements for the **Smart SOS Device** could include integrating AI-driven analytics to predict and prevent potential emergencies by monitoring user behavior and patterns. Advanced communication modules, such as satellite connectivity, could be added to ensure functionality in remote areas without network coverage.

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